SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	**ITA05**	**Course Title:**	**Computer Vision**
CO Assessed:	**CO4 – Articulation (combining methods for evaluation) P4**	**Assessment:**	**Assessment Tool 1 – Data Interpretation**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO4 Statement:	*Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)*		
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Section / Batch:	ITA0512	**Date:**	17/08/2026

Code of Conduct

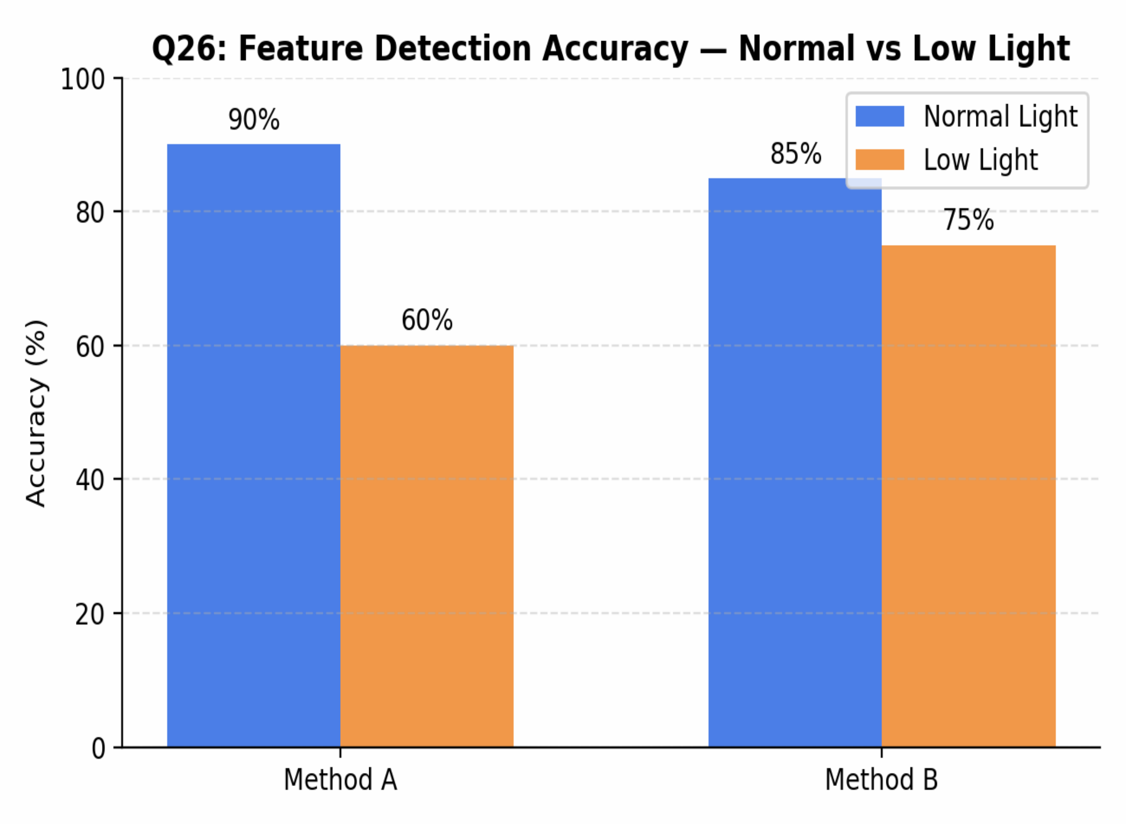
I Yavanika S [192521258] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Yavanika S

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

➢ Question 26: Feature Detection under Illumination Changes

Method	Normal Light	Low Light
A	90%	60%
B	85%	75%



Understanding of Data

Method A achieves higher accuracy under normal lighting (90%) but drops sharply to 60% in low light, a fall of 30 percentage points. Method B starts slightly lower under normal conditions (85%) but retains 75% accuracy in low light, a much smaller drop of only 10 points.

Analysis

- Method A is optimized for well-lit, controlled conditions and relies heavily on strong gradient/edge information that degrades when illumination is poor.
- Method B likely uses illumination-invariant features (e.g., normalized gradients, histogram equalization, or adaptive thresholding), making it far more stable across lighting conditions.
- The performance gap ($\Delta = 30\%$ for A vs $\Delta = 10\%$ for B) is the key robustness indicator, not the peak accuracy alone.

Application of Concepts

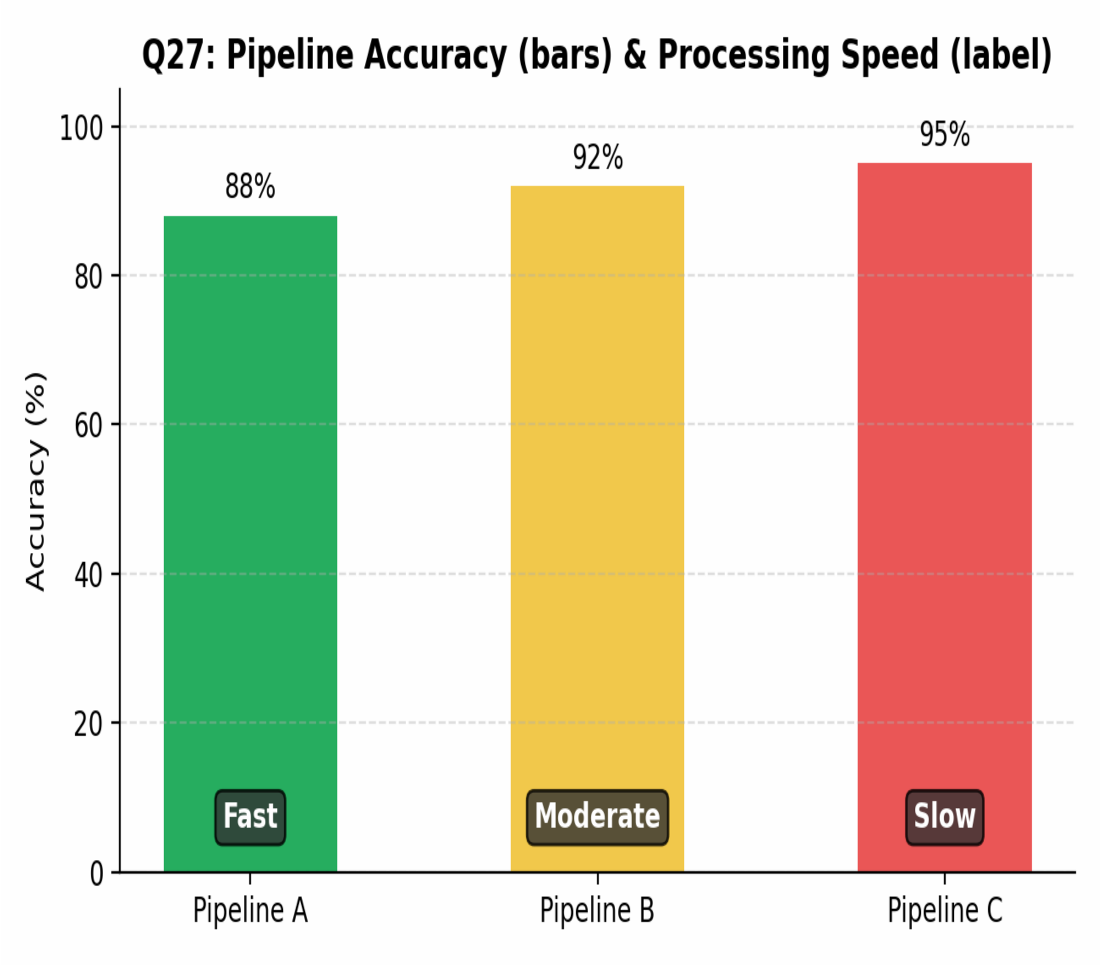
Robustness in feature detection is judged by consistency across environmental variation, not just peak-condition accuracy. A small performance drop under adverse conditions indicates a more generalizable and reliable feature descriptor.

Conclusion

Method B is the more robust technique overall. Although Method A performs marginally better in ideal lighting, Method B's stability across normal and low-light conditions makes it the safer choice for real-world deployments such as outdoor surveillance, night-time driving assistance, or any environment with unpredictable lighting.

Question 27: Video Processing Pipeline Comparison

Pipeline	Accuracy	Speed
A	88%	Fast
B	92%	Moderate
C	95%	Slow



Understanding of Data

The three pipelines show a clear inverse relationship between accuracy and speed: Pipeline A is fastest but least accurate (88%), Pipeline C is most accurate (95%) but slowest, and Pipeline B sits in between on both dimensions (92%, Moderate speed).

Analysis

- Pipeline A likely uses lightweight models or frame skipping, sacrificing accuracy for throughput.
- Pipeline C likely applies deeper networks or processes every frame at full resolution, increasing accuracy at the cost of computation time.
- Pipeline B represents a balanced engineering trade-off between the two extremes.

Application of Concepts

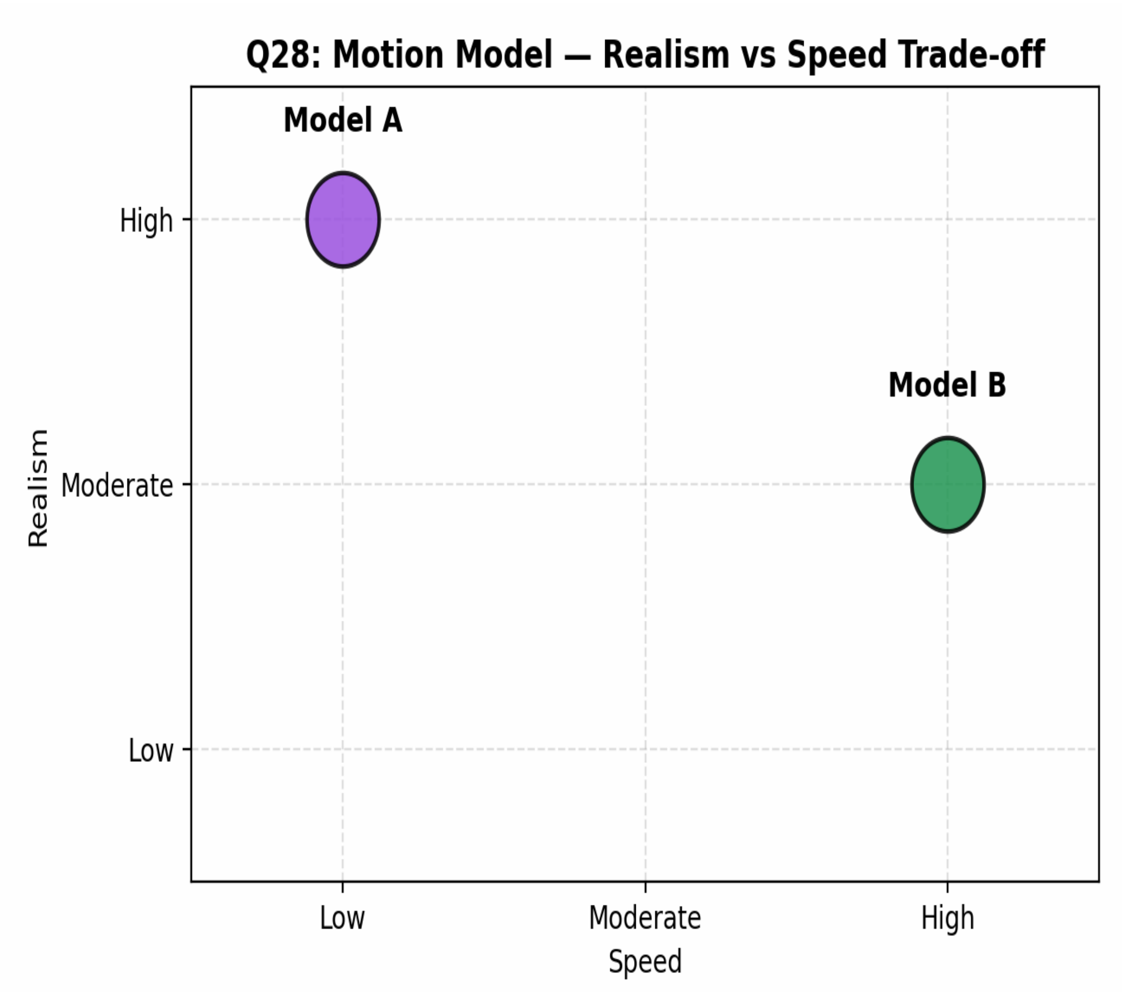
Suitability depends entirely on the target use case: real-time systems (e.g., live video conferencing, traffic monitoring) demand speed even at reduced accuracy, whereas offline or safety-critical analysis (e.g., medical video review, forensic analysis) can tolerate slower processing for higher accuracy.

Conclusion

There is no single 'best' pipeline in isolation. Pipeline A suits real-time, latency-sensitive applications; Pipeline C suits offline, accuracy-critical applications; and Pipeline B is the recommended default for general-purpose systems that need a reasonable balance of both speed and accuracy.

Question 28: Motion Model Evaluation

Model	Realism	Speed
A	High	Low
B	Moderate	High



Understanding of Data

Model A produces highly realistic motion but processes it slowly, while Model B produces moderately realistic motion but runs at high speed — a direct trade-off between visual fidelity and computational efficiency.

Analysis

- Model A likely uses physics-based or high-order motion equations that better capture natural, non-linear movement, at the cost of computation.
- Model B likely uses a simplified motion model (e.g., linear or constant-velocity assumptions), which sacrifices some realism for speed.
- The choice depends on whether the application values visual/behavioral accuracy or responsiveness more.

Application of Concepts

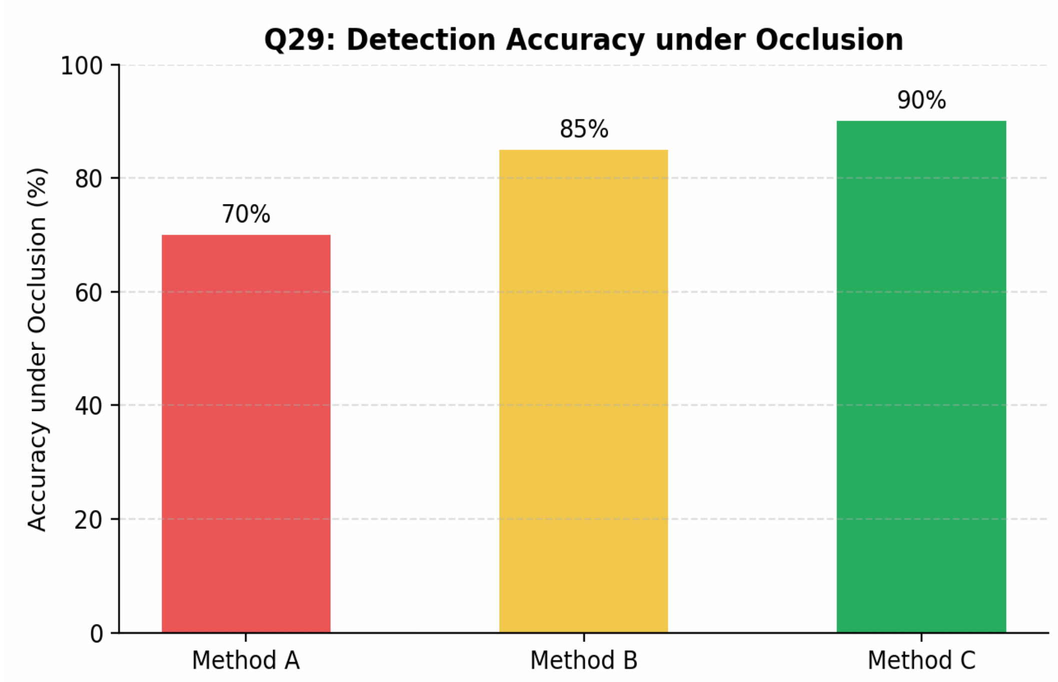
In application-critical contexts, the deciding factor is the cost of an error versus the cost of delay. High-realism motion models matter where physical accuracy drives decisions; high-speed models matter where responsiveness or scale is the priority.

Conclusion

Model A is best suited for applications such as animation, simulation, or biomechanical analysis where realism is critical and processing can happen offline. Model B is best suited for real-time applications such as gaming, robotics, or live tracking, where responsiveness outweighs the need for perfect realism.

Question 29: Detection under Occlusion

Method	Accuracy
A	70%
B	85%
C	90%



Understanding of Data

Accuracy under occlusion increases progressively from Method A (70%) to Method B (85%) to Method C (90%), indicating a clear ranking in robustness to partially hidden or blocked objects.

Analysis

- Method A likely relies on complete object appearance and struggles when key features are hidden.
- Method B likely incorporates partial-feature matching or contextual cues, improving performance when parts of the object are missing.
- Method C likely uses strong context modeling (e.g., part-based models, attention mechanisms, or temporal continuity in video) to infer occluded regions, giving it the best robustness.

Application of Concepts

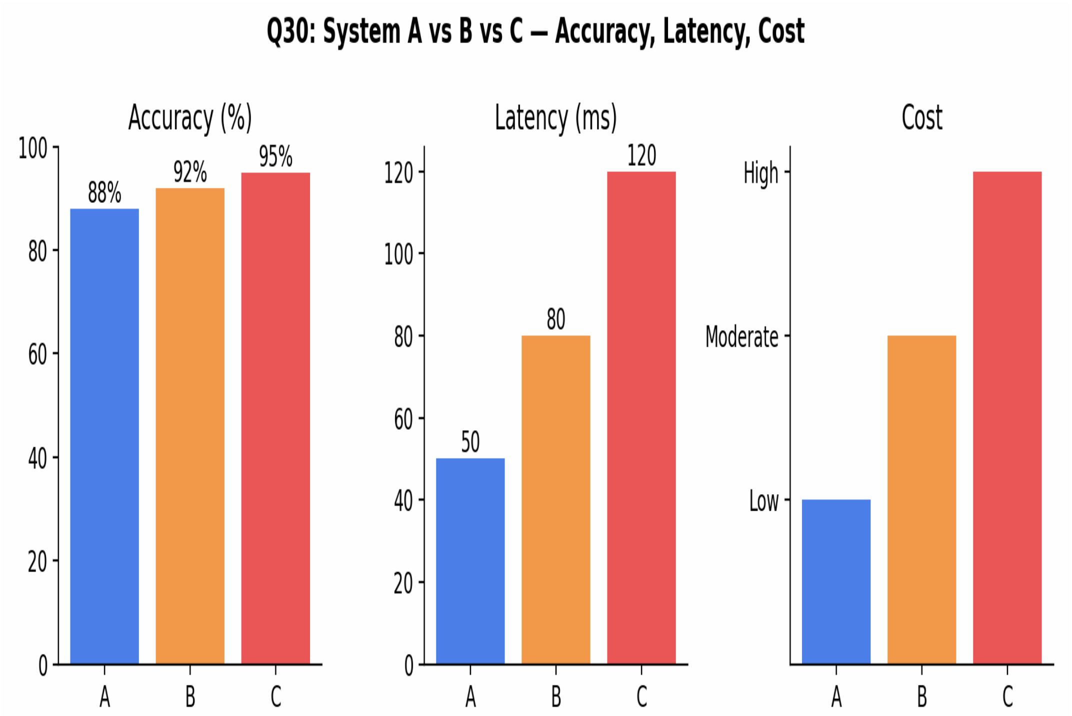
Occlusion handling is critical in real-world environments such as crowded scenes, autonomous driving, and surveillance, where objects are frequently and unpredictably blocked by other objects or people.

Conclusion

Method C is the most robust technique under occlusion and is recommended for applications with frequent partial visibility, such as pedestrian detection in traffic or object tracking in crowded scenes. Method A is only advisable in controlled environments where occlusion is rare.

Question 30: System Performance under Constraints

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High



Understanding of Data

As accuracy increases from Model A (88%) to Model C (95%), both latency and cost increase proportionally — latency rises from 50 ms to 120 ms, and cost rises from Low to High. This shows a classic three-way trade-off among accuracy, speed, and cost.

Analysis

- Model A offers the best efficiency (fastest, cheapest) but the lowest accuracy of the three.
- Model C offers the best accuracy but is more than twice as slow as Model A and carries the highest cost.
- Model B offers a balanced middle ground, gaining 4% accuracy over A for a moderate increase in latency and cost.

Application of Concepts

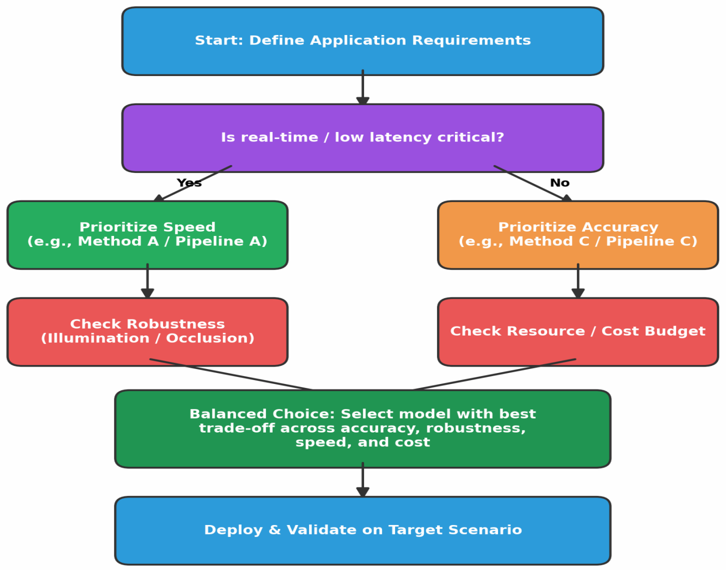
Choosing the 'best' system requires weighing constraints against the deployment context: edge devices and real-time systems with tight budgets favor low latency and cost; high-stakes applications (e.g., medical diagnosis, security-critical detection) can justify higher latency and cost for maximum accuracy.

Conclusion

For resource-constrained or real-time deployments (e.g., mobile or embedded vision systems), Model A is the best choice. For accuracy-critical, less time-sensitive deployments (e.g., medical imaging analysis), Model C is preferable. For most general-purpose production systems, Model B offers the best overall balance of accuracy, latency, and cost, and is the recommended default.

Decision Framework: How to Choose the Right Model

The flowchart below summarizes the general decision process applied across all five questions — matching technique, pipeline, or model to the demands of the target application.



Overall Summary

Across all five comparisons, a consistent pattern emerges: methods optimized for raw accuracy tend to be slower and less robust to real-world variation (illumination, occlusion), while methods optimized for speed and efficiency tend to sacrifice some accuracy or robustness.